



Examiners' Report Lead Examiner Feedback

January 2024

Pearson BTEC Nationals in
Applied Science/Forensic and Criminal
Investigation (31617H)
Unit 1: Principles and Applications of Science I

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this [link](#).

Introduction

Biology

Medical professionals need to understand the structure and workings of cells. They build on this knowledge to understand how the body stays healthy as well as the symptoms and causes of some diseases. This allows them to diagnose and treat illnesses. The study of bacterial prokaryotic cells gives an understanding of how some other diseases are caused and can be treated.

Chemistry

This paper covers key chemistry concepts including atomic structure and bonding and chemical and physical properties of substances to their uses.

Physics

The examination was designed to allow learners to show their knowledge and understanding of 'Waves in Communication'. The examination was comparable to those set previously as there were no changes to the specification or the composition of the paper.

The topics included longitudinal and transverse waves and the application of these waves. The applications included transmission of signals by satellite communication systems, the structure of an optical fibre, conversion of analogue signals to digital and the passage of light waves through a diffraction grating.

Overall Performance of the Unit

Biology

Learners should show that they understand the relationship between the structure and function of cells and tissues throughout the unit. Learners still struggle to recall definitions and correctly evaluate appropriate calculations, such as a percentage change.

Learners often lost marks because they did not understand the requirement of the command verb, particularly "explain", where an identification point with a linked expansion to give an explanation is

required to gain the marks. For other questions, some learners tried to explain when they only needed to “describe” or “state”.

Most learners were able to access the extended open response question and gain at least a mark in level 1 for their knowledge of the specialisation of the root hair cell.

Chemistry

Learners that were successful this year showed a good understanding of key terms, scientific vocabulary and mathematical skills and were able to apply this knowledge and understanding to new scenarios. They were precise with their language and explained in detail where necessary.

Learners that performed well, carried out their calculations in a clear and logical manner which helped them to arrive at the correct answer and were able link ideas together.

Physics

Learners were generally able to identify an area of compression in a longitudinal wave and were able to recall areas of the electromagnetic spectrum.

Describing the motion of coils in a spring as a longitudinal wave passes through was found to be more difficult.

Learners were able to demonstrate their ability in completing calculations, although some learners still had difficulty in correctly rearranging equations.

Knowledge of the difference between upload and download signals in a satellite communication system was not clearly understood.

Although optical fibres are in everyday use, understanding the purpose of glass cladding in their design was limited. Similarly, many learners struggled to explain the process of conversion of analogue signals to digital ones and the reasoning behind the regular regeneration of digital signals when sent over a long distance.

The production of a pattern of light and dark regions on a screen following the passage of laser light through a diffraction grating required learners to have knowledge of the behaviour of light waves as they pass through slits,

and of the concepts of constructive and destructive interference. Some learners were able to link these concepts to the production of the light and dark regions and a minority were able to demonstrate deeper understanding by explaining why constructive and destructive interference occurs. However, the majority of learners found it difficult to describe what happened to the light once it had passed through the diffraction grating.

Chemistry

Question 1

This question focused on carbon and some of its compounds.

1(a)(ii) asked learners to complete the electronic configuration for carbon. In general, many learners understood the question and were able to give the correct answer using the correct scientific notation. Those that did not gain mark often knew that two electrons were present but lost the mark as they gave incorrect shells or quantum numbers.

A common incorrect response was $3s^2$, as in this example that scored no marks.

(ii) Complete the electronic configuration for carbon.

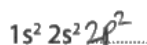
(1)



This response scored one mark.

(ii) Complete the electronic configuration for carbon.

(1)



1(b)(i) asked learners to give a use of carbon dioxide. Learners found this more challenging than Q1(a)(ii), with a lower percentage gaining the mark and many giving a process that *produced* carbon dioxide, such as burning fossil fuels, rather than a use.

This response gained no marks.

(b) Carbon dioxide and methane are compounds containing carbon.

(i) State **one** use of carbon dioxide.

(1)

Burning fossil fuels

Where learners did score the mark, it was often for stating its use of promoting photosynthesis. Photosynthesis alone scored the mark, as in this case.

(b) Carbon dioxide and methane are compounds containing carbon.

(i) State **one** use of carbon dioxide.

(1)

photosynthesis

In some cases, learners were too vague with their answers and just stated that it was used in agriculture or for plant growth, this was insufficient for credit and did not gain credit.

(b) Carbon dioxide and methane are compounds containing carbon.

(i) State **one** use of carbon dioxide.

(1)

Plant growth

(b) Carbon dioxide and methane are compounds containing carbon.

(i) State **one** use of carbon dioxide.

(1)

Agriculture

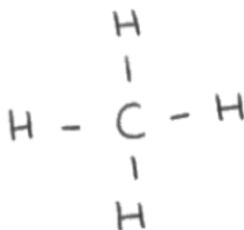
1(b)(ii) asked learners to draw the 3D line diagram for methane. Learners found this very hard with the majority drawing a simple 2D diagram that they have been used to seeing at level 2.

The following response gained no marks.

Methane has the formula CH_4 .

(ii) Draw the 3D line diagram for methane.

(1)

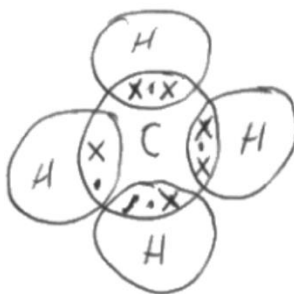


Some learners did not read the question carefully and gave a dot and cross diagram rather than the 3D line diagram. This also scored no marks.

Methane has the formula CH_4 .

(ii) Draw the 3D line diagram for methane.

(1)

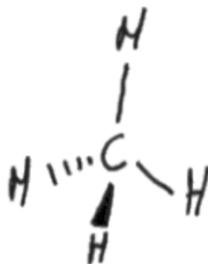


Those that did score often draw a fully correct 3D diagram, using dotted lines and wedges appropriately, as in this example which gained credit. Those that drew single lines, without wedges also gained credit for showing the correct tetrahedral shape.

Methane has the formula CH_4 .

(ii) Draw the 3D line diagram for methane.

(1)



1(b)(iii) asked gave learners a table showing the bond energies of a C=O and a C-H bond and asked learners to explain why the C=O is stronger than the C-H bond. A good proportion of learners were able to access this question and understood that the C=O bond is stronger than the C-H bond because it is a double bond and that it would require more energy to break that bond. Fewer however were able to explain this in terms of the number of shared pairs of electrons or that these would mean that there is a stronger attraction between these shared pairs and the nuclei of the atom.

This response demonstrates a good answer which scored the four marks available.

Table 1 shows the bond energy of a C—H bond in methane and a C=O bond in carbon dioxide.

bond	bond energy (kJ/mol)
C—H	414
C=O	805

Table 1

(iii) Using the information in Table 1, explain why the C=O bond is stronger than the C—H bond.

(4)

- double bonds are stronger than single bonds.
because they share two pairs of electrons,
Stronger attraction with the positive nucleus.
~~Stronger electron~~ Requires more energy to overcome
or break down.

- because it is a double bond, it is stronger
- more electrons in the C=O so it has more attraction with
the positive nucleus require more energy to overcome.

(Total for Question 1 = 8 marks)

Question 2

This question focused on the Halogens.

2(a)(ii) gave learners a table showing the boiling points of the first four halogens and asked them to explain the trend shown. A good proportion of learners were able to gain the first mark for recognising that the boiling points of the elements increased down the group, of these around half were then able to start to explain this in terms of there being more electrons or more shells of electrons. Fewer stated that the atoms became larger. A small number of learners were then able to link this to the fact there were more van der Waals forces.

In many cases, learners thought that to boil the halogens, the atoms needed to be separated as in this example.

This response scored one mark for the identification of the trend.

Table 2 shows the boiling points of four halogens.

element	boiling point (°C)
fluorine	-188
chlorine	-35
bromine	59
iodine	184

Table 2

(ii) Explain the trend in the boiling points of the halogens in Table 2.

(3)

As you go down group 7, the boiling point increases. This is because bonds between the atoms require more heat energy to break further down the ~~table~~ group as the forces of attraction are stronger

The following response shows a good example that scored all three marks.

Table 2 shows the boiling points of four halogens.

element	boiling point (°C)
fluorine	-188
chlorine	-35
bromine	59
iodine	184

Table 2

(ii) Explain the trend in the boiling points of the halogens in Table 2.

(3)

boiling point increases down the group
this is because the elements gain electrons,
increasing the London dispersion forces
so more energy is required to ^{break} overcome the
inter molecular forces.

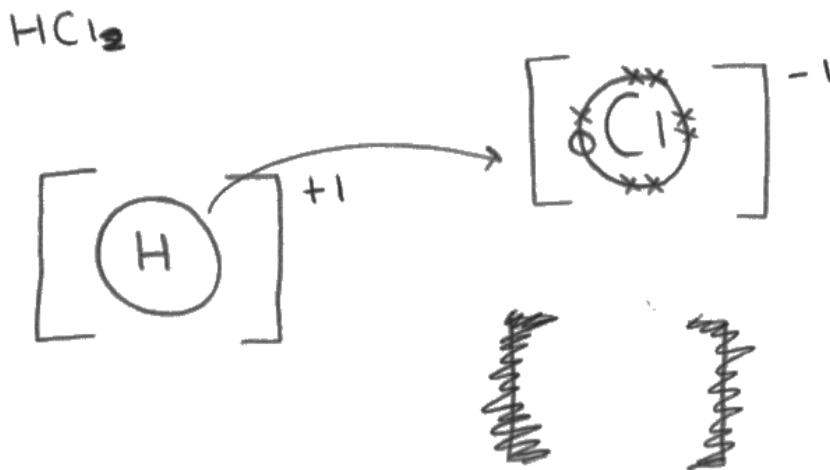
2(b)(i) asked learners to draw the dot and cross diagram for a molecule of hydrogen chloride. A good proportion of learners showed one pair of electrons, but many added an extra electron on the hydrogen atom or an incorrect number of electrons on the chlorine atom. Often learners added extra atoms into the diagram or thought that the atoms were bonded ionically so gained no marks.

This response demonstrates a common incorrect answer that scored no marks.

(b) Chlorine reacts with hydrogen to form hydrogen chloride.

(i) Draw the dot and cross diagram for a molecule of hydrogen chloride.

(2)

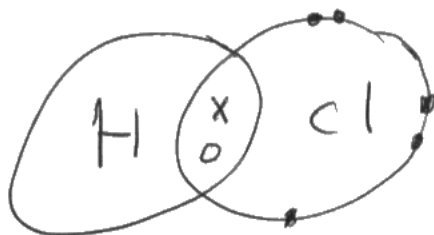


This response scored one mark for the correct shared pair of electrons but lost the second mark as the number of electrons on the chlorine is incorrect.

(b) Chlorine reacts with hydrogen to form hydrogen chloride.

(i) Draw the dot and cross diagram for a molecule of hydrogen chloride.

(2)

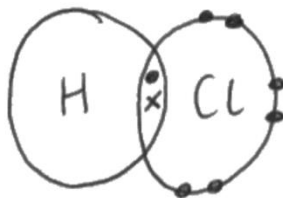
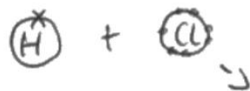


The following response is a good example which scored both marks available.

(b) Chlorine reacts with hydrogen to form hydrogen chloride.

(i) Draw the dot and cross diagram for a molecule of hydrogen chloride.

(2)



2(b)(ii) asked learners to explain why a molecule of hydrogen chloride is polar.

Learners found the question challenging with many learners leaving the answer blank and only a small proportion gaining both marks for understanding that there is a large difference in negativity between the hydrogen and the chlorine and that this results in a partial positive charge on the hydrogen and a partial negative charge on the chlorine.

Where learners scored, it was often for drawing a diagram showing the charges. Some also showed the electron pair closer to the chlorine on their diagram as in the following response, which demonstrated a good understanding and scored both marks.

(ii) Explain why a molecule of hydrogen chloride is polar.

You may draw a diagram to support your answer.

(2)



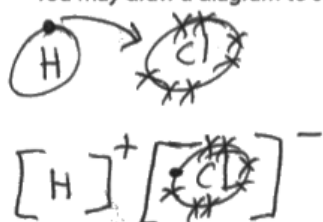
A molecule of hydrogen chloride is polar the electrons are more attracted to the negative side which is the chlorine. This means that the molecule is more electronegative as the electrons are not equal and right in the middle.

In some cases, learners thought that the elements formed an ionic bond and so lost both marks, as in this example.

(ii) Explain why a molecule of hydrogen chloride is polar.

You may draw a diagram to support your answer.

(2)



polar is when an element gains electrons and the other loses them. to gain a full outer shell due to ionic bonding. H has lost 1 electron (e^-) and became a $+1$ charge, Cl has gained this e^- and became $1-$ charge. Both elements now have a full outer shell.

(Total for Question 2 = 8 marks)

Question 3

This question focused on iron and iron oxide.

3(b) asked learners for the oxidation number of iron in iron and in iron oxide.

It was clear that learners did not have a clear understanding of the term oxidation number, with many giving the mass number of iron and the mass of iron in iron oxide rather than the oxidation numbers as in this response which scored no marks.

Iron can be extracted from iron oxide.

(b) Give the oxidation number of iron in Fe and in Fe_2O_3 .

(2)

oxidation number of iron in Fe = 55.8

oxidation number of iron in Fe_2O_3 = 111.6

The following response demonstrates a good answer that scored both marks.

Iron can be extracted from iron oxide.

(b) Give the oxidation number of iron in Fe and in Fe_2O_3 .

(2)

oxidation number of iron in Fe = 0

oxidation number of iron in Fe_2O_3 = 3

3(c)(i) asked learners to calculate the minimum mass of iron oxide that was used to form 27.90g of iron in a reaction.

Learners found this quite difficult with a small proportion of learners being able to give the correct answer of 39.9 to gain full marks.

A good proportion were able to calculate the number of moles of iron formed, but often forgot to divide this number by 2 before calculating the mass therefore gaining for coming to answer of 79.8 rather than the full three marks.

In the following example, the learner scored one mark for calculating the number of moles correctly.

(c) A sample of iron oxide was reduced with excess carbon monoxide.

The theoretical yield of iron from this sample is 27.90 g.



(i) Calculate the minimum mass of iron oxide that was used.

(relative atomic mass: Fe = 55.8, relative formula mass: $\text{Fe}_2\text{O}_3 = 159.6$)

(3)

Show your working.

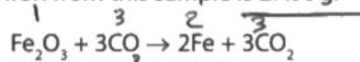
$$\frac{27.90}{55.8} = 0.5$$

minimum mass of iron oxide = g

In the following response the learner has incorporated calculating the number of moles into their first step, multiplying the mass of iron by 2, but then divided the answer by 2 again in step 2 so arrived at an answer of 19.95 rather than 39.9. This was a common error.

(c) A sample of iron oxide was reduced with excess carbon monoxide.

The theoretical yield of iron from this sample is 27.90 g.



(i) Calculate the minimum mass of iron oxide that was used.

(relative atomic mass: Fe = 55.8, relative formula mass: $\text{Fe}_2\text{O}_3 = 159.6$)

(3)

moles = $\frac{\text{mass}}{\text{Mr}}$

Show your working.

$$\frac{27.90}{111.6} = 0.25$$

$$0.125 = \frac{\text{mass}}{159.6}$$

$$\begin{array}{ccccccc} 1 & : & 3 & : & 2 & : & 3 \\ 0.125 & : & 0.75 & : & 0.25 & : & 0.75 \end{array}$$

$$0.125 \times 159.6 = 19.95$$

minimum mass of iron oxide = 19.95 g

In this response, the learner has correctly calculated the number of moles of iron. At this point the learner has incorrectly multiplied their answer by 2 rather than divided and then used this answer to find the mass of the iron oxide.

This response gained two marks and demonstrates the importance of showing working in an ordered way so that the ECF marks could be awarded, as an answer of 159.6 on the answer line with no working would have gained no credit.

(c) A sample of iron oxide was reduced with excess carbon monoxide.

The theoretical yield of iron from this sample is 27.90 g.



(i) Calculate the minimum mass of iron oxide that was used.

(relative atomic mass: Fe = 55.8, relative formula mass: Fe_2O_3 = 159.6)

(3)

Show your working.

Handwritten working:

Diagram: A triangle with 'mass' at the top, 'moles' at the bottom left, and 'RFM' at the bottom right.

	Fe_2O_3	Fe
mass	?	27.90
RFM	159.6	55.8
moles	1	0.5

Below the table, the calculation $159.6 \times 1 = 159.6$ is shown.

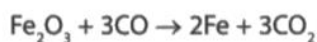
Below the table, the calculation $\frac{27.90}{55.8}$ is shown.

minimum mass of iron oxide = 159.6 g

The next response scored the full three marks.

(c) A sample of iron oxide was reduced with excess carbon monoxide.

The theoretical yield of iron from this sample is 27.90 g.



(i) Calculate the minimum mass of iron oxide that was used.

(relative atomic mass: Fe = 55.8, relative formula mass: $\text{Fe}_2\text{O}_3 = 159.6$)

ratio = 1 : 2 (3)

Show your working.

$$\text{Moles} = \frac{\text{Mass}}{M_r} \quad x = \frac{27.90}{55.8}$$

$$\begin{aligned} \text{Mass} &= \text{Moles} \times M_r \\ x &= 0.25 \times 159.6 \\ x &= 39.9 \end{aligned}$$

$$\begin{aligned} x &= 0.5 \\ 1 : 2 \\ 0.25 : 0.5 \end{aligned}$$

minimum mass of iron oxide = 39.9 g

3(c)(ii) Learners were provided with the actual yield for the reaction and asked to calculate the percentage yield. It was pleasing to see that the majority of learners were able to use the equation provided to calculate the percentage yield correctly, as in this response, which scored two marks.

The actual yield of iron produced from this sample was 19.53 g.

(ii) Calculate the percentage yield of iron.

(2)

Use the equation

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$\frac{19.53}{27.90} \times 100 = 70\%$$

percentage yield of iron = 70 %

In some cases, the learners forgot to multiply their number by 100 and gained just one mark.

The actual yield of iron produced from this sample was 19.53 g.

(ii) Calculate the percentage yield of iron.

(2)

Use the equation

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

~~20/10/20~~

$$\frac{19.53}{27.90} = 0.7$$

percentage yield of iron = 0.7 %

Question 4

This was the last question on the paper with up to six marks available, though a level-based mark scheme. Learners were provided with a set of observations of the reactions of lithium, sodium and potassium and water. They were asked to explain the similarities and differences between the three reactions.

At Pass level, learners were expected to analyse the question and describe the similarities and differences between the three reactions.

At Merit level, learners were expected to explain these differences and they often concentrated on the reactivity of the metals.

At Distinction level, learners were expected to provide more depth and justification to explanations in similarities and differences.

In the following response, the learner has described similarities and differences the observations. Unfortunately, they have not attempted to explain any of these observations, gaining two marks in Level 1.

4 A teacher reacted a small piece of lithium with water containing universal indicator.

The teacher repeated the reaction using sodium and potassium, using fresh water and universal indicator each time.

A learner observed the reactions and recorded the results in Table 3.

metal	observation
lithium	- fizzing - metal floats and moves around the surface of the water - universal indicator turns purple
sodium	- rapid fizzing - metal floats and moves quickly around the surface of the water - universal indicator turns purple
potassium	- very rapid fizzing - lilac flame and popping sound - metal floats and moves very quickly around the surface of the water - universal indicator turns purple

Table 3

Explain the similarities and differences between these three reactions.

(6)

One similarity of these three reactions is fizzing ~~although the amount~~ even though there is a similarity in the fizzing there is also a difference in all three of the reactions fizz at different speeds. Another similarity of all three of the reactions is that the universal indicator turns purple with all of the reactions. Potassium is the only reaction that makes a flame and a popping sound. All three of the metals being lithium, sodium and potassium are all in the s-block and in Group 1.

Although all the metals are in group 1 they all have different atomic numbers and relative atomic mass. One last similarity and difference of all three of the metals is that they all float on water but move at different speeds.

(Total for Question 4 = 6 marks)

In the following response, the learner has given a similarity and some differences. They have given one very basic explanation – that the differences are because potassium is most reactive. This was just sufficient for marks three marks in Level 2.

- 4 A teacher reacted a small piece of lithium with water containing universal indicator.

The teacher repeated the reaction using sodium and potassium, using fresh water and universal indicator each time.

A learner observed the reactions and recorded the results in Table 3.

metal	observation
lithium	- fizzing - metal floats and moves around the surface of the water - universal indicator turns purple
sodium	- rapid fizzing - metal floats and moves quickly around the surface of the water - universal indicator turns purple
potassium	- very rapid fizzing - lilac flame and popping sound - metal floats and moves very quickly around the surface of the water - universal indicator turns purple

Table 3

Explain the similarities and differences between these three reactions.

(6)

lithium had abit of jizing, sodium had rapid fizzing and potassium had very rapid fizzing this show that potassium was the most reactive. ~~they all~~ the universal indicator turned ~~for~~ purple for all of them. lithium only moves around the surface of water a little bit and then sodium moves quickly and potassium moves very quickly which can also indicate to us that it was the most reactive. potassium

was the only one who has a lilac flame and a popping sound. they each reacted similarly however potassium reacted the most and was the only one who had a lilac flame with a popping sound. We can see that as you go down group 1 they get more reactive.

(Total for Question 4 = 6 marks)

In the following response, the learner has described similarities and differences and given some basic explanations, that the fizzing is because hydrogen is produced, and it is faster because they are more reactive. This was sufficient for four marks in Level 2.

4 A teacher reacted a small piece of lithium with water containing universal indicator.

The teacher repeated the reaction using sodium and potassium, using fresh water and universal indicator each time.

A learner observed the reactions and recorded the results in Table 3.

metal	observation
lithium	- fizzing - metal floats and moves around the surface of the water - universal indicator turns purple
sodium	- rapid fizzing - metal floats and moves quickly around the surface of the water - universal indicator turns purple
potassium	- very rapid fizzing - lilac flame and popping sound - metal floats and moves very quickly around the surface of the water - universal indicator turns purple

Table 3

Explain the similarities and differences between these three reactions.

(6)

~~Similar~~ The similarities in this reaction are that the metal floats ^{and moves} around the surface of the water. This is because group 1 metals are insoluble and don't dissolve in water. Another similarity is that the universal indicator turns purple. This is because group 1 metals are alkali. The final similarity is that the metals fizz when added to water. This is because group 1

metals are ^{being} reactive and they produce hydrogen. The differences in this experiment are that the further the teacher goes down group 1, the ~~more~~ faster the fizzing or the metal. This is because as you go down group 1, the metals become more reactive and Potassium has very rapid fizzing as ~~the~~ ^{more} electrons are delocalising into the water. Another difference is that Potassium metal ~~travels~~ moves ~~to~~ quicker than sodium and sodium moves faster than potassium around the surfaces of the water. This is because the more electrons the metal has, the faster it moves on the surface of the water so Potassium moves the fastest.

(Total for Question 4 = 6 marks)

In the following response, the learner has given similarities and differences between the reactions and explained three of them with two with detail so a mark of six in Level 3 was awarded.

4 A teacher reacted a small piece of lithium with water containing universal indicator.

The teacher repeated the reaction using sodium and potassium, using fresh water and universal indicator each time.

A learner observed the reactions and recorded the results in Table 3.

metal	observation
lithium	- fizzing - metal floats and moves around the surface of the water - universal indicator turns purple
sodium	- rapid fizzing - metal floats and moves quickly around the surface of the water - universal indicator turns purple
potassium	- very rapid fizzing - lilac flame and popping sound - metal floats and moves very quickly around the surface of the water - universal indicator turns purple

metal + water $\rightarrow$ ~~hydroxide~~ ^{metal} **Table 3**

Explain the similarities and differences between these three reactions.

- ~~At~~ There is fizzing in all 3 reactions ^{because the metal is (6) reacting to the water.}
- The metal floats to the surface in all 3 reactions because the metals have a lower density than the water.
- The universal indicator turns purple in all 3 reactions because the metal oxide produced each time is the same pH.
- Potassium has the most rapid fizzing because it is the most reactive metal out of all 3.
- Lithium has the least rapid fizzing because it is the least reactive of all 3.
- An ionic compound is formed in all 3 reactions. A metal oxide is formed.
- Potassium moves the quickest around the surface of water.

because it had the fastest reaction time.

- Sodium is more reactive than Lithium and so had more rapid fizzing than Lithium when reacting with water.
- Potassium is the most reactive metal out of all 3 because it has the largest atom. So the electrostatic attraction between the outer electron and nucleus is weaker, making it easier to lose the single electron.

(Total for Question 4 = 6 marks)

TOTAL FOR PAPER = 30 MARKS